

SmartInterestX — Loan & Interest Management

Mobile Application

Project Development Guide

Project Overview

SmartInterestX is a fintech-style mobile application designed to help users track, calculate, and manage interest-based loan transactions between borrowers and lenders. The application simplifies financial tracking by automating loan calculations, payment tracking, reminders, and analytics reporting.

Key Features

- Loan and interest calculations
- Due date tracking
- Payment proof management
- Push notifications for reminders
- Analytics dashboard for interest earned or paid

Technology Stack

Layer	Technology Used
Frontend	Flutter (Dart)
Backend	Firebase
Authentication	Firebase Authentication
Database	Cloud Firestore
File Storage	Firebase Storage
Notifications	Firebase Cloud Messaging

Application Workflow

- User signs up or logs in using Firebase Authentication.
- User accesses the dashboard showing loan summary.
- User can add new loans with borrower details.
- Interest is calculated automatically.
- User tracks payments and uploads payment proof.
- Application sends due-date reminders via notifications.
- Analytics dashboard shows total interest and payments.

Project Setup Steps

- Install Flutter SDK and verify using 'flutter doctor'.
- Install VS Code with Flutter and Dart extensions.
- Install Android Studio for emulator support.
- Create Flutter project using 'flutter create smart_interest_x'.
- Navigate into the project folder and open in VS Code.
- Run the project using 'flutter run'.

Required Flutter Packages

- firebase_core
- firebase_auth
- cloud_firestore
- firebase_storage
- image_picker
- intl
- flutter_local_notifications
- provider

Recommended Project Folder Structure

```
lib/  
  screens/  
    login_screen.dart  
    signup_screen.dart  
    dashboard_screen.dart  
    add_loan_screen.dart  
    loan_detail_screen.dart  
  models/  
    loan_model.dart  
  services/  
    auth_service.dart  
    firestore_service.dart  
    notification_service.dart  
  widgets/  
    loan_card.dart  
    custom_button.dart  
  utils/  
    interest_calculator.dart  
  main.dart
```

Interest Calculation Formula

Simple Interest Formula:

$\text{Interest} = (\text{Principal} \times \text{Rate} \times \text{Time}) / 100$

Example:

Principal = 10,000

Rate = 10%

Time = 1 year

Interest = 1,000

Development Roadmap

- Week 1: Project setup, Firebase integration, Authentication
- Week 2: Dashboard UI and Loan management screens
- Week 3: Payment tracking and proof upload
- Week 4: Notifications, analytics dashboard, testing and deployment

Conclusion

SmartInterestX provides students with a complete real-world fintech application development experience. The project demonstrates Flutter mobile development, Firebase backend integration, data management, notifications, and financial calculation logic. By completing this project, students gain hands-on experience building scalable production-level applications.